Speech Emotion Recognition System

基於 1D CNN 和 MFCC 的語音情感識別

2025/12/23

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01_

Introduction

論文簡介

Speech Emotion Recognition (SER)

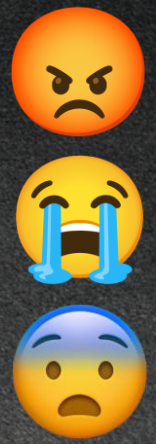
- 讓電腦透過語音識別人類情感

滿腦子就只想到自己呢

從來不覺得玩樂團開心過

只要是我能做的，我什麼都願意做

→



Paper Highlight

- 傳統機器學習太笨，準確率低
- 深度學習有好結果，但資料不夠就會 overfitting
- 用 1D CNN 加上 MFCC，再做資料增強
- 用更少的資源，達到更高的準確率

Evaluation

- Accuracy、Recall、Precision、F1-Score、Confusion Matrix
- 在 RAVDESS 資料集達到 97.33% 的準確率
  - 由 24 位專業演員在錄音室環境下錄製，標準北美口音
- 在 CREMA-D 資料集達到 94.69% 的準確率
  - 由 91 位演員錄製，說話者多樣性

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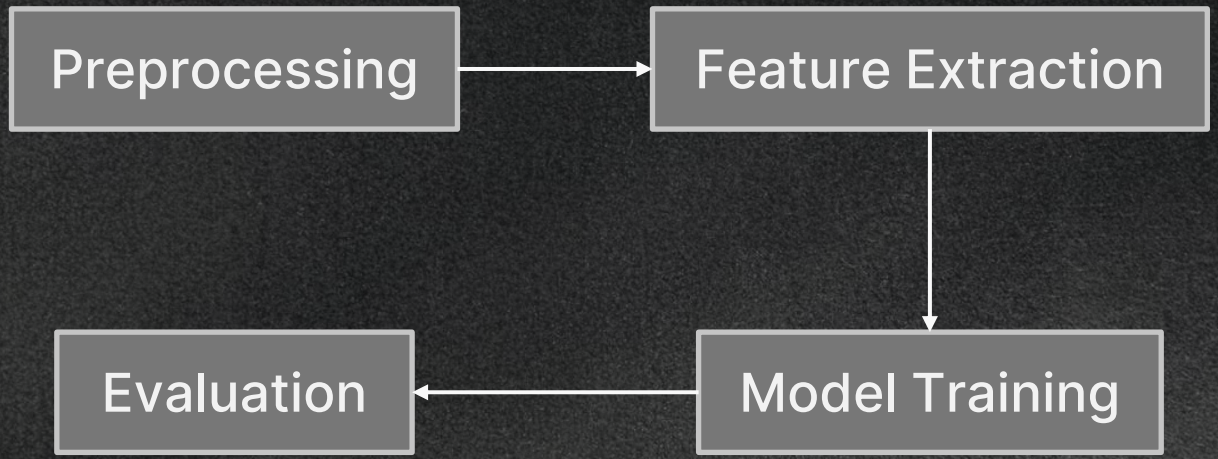
Methodology

程式實作

The Big Picture



Preprocessing

- 標準化：把音檔統一成 2.5 秒
- 靜音去除
- 資料增強：解決資料不足的問題

Preprocessing - 22050 Hz

- Nyquist-Shannon Sampling Theorem

Theorem—If a function $x(t)$ contains no frequencies higher than B hertz, then it can be completely determined from its sampled values at a sequence of points spaced less than $1/(2B)$ seconds apart.

定理 —— 如果一個函數 $x(t)$ 中沒有高於 B 赫茲的頻率，那麼它可以完全從一串間距不到 $1/(2B)$ 秒的點取樣值中確定。

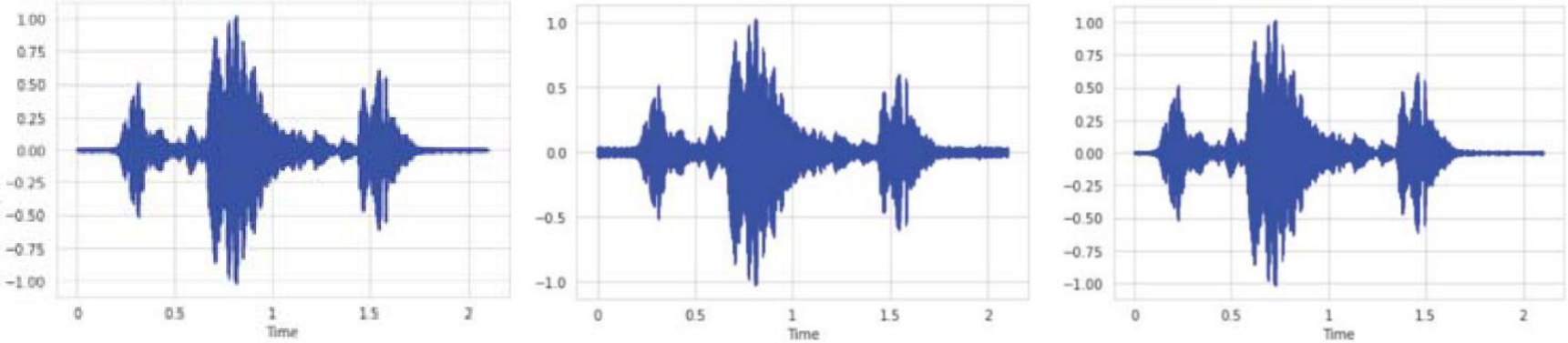
- 11,025 Hz 的帶寬足以完整保留人類語音中所有的情緒細節

Silence Trimming

- 音檔開頭沒聲音
- 切掉前面 0.5 秒
- 簡單不耗效能

```
# original data
data, sample_rate = librosa.load(file_path, duration=2.5, offset=0.5, sr=22050)
```

Data Augmentation



(a) Original speech waveform (b) The waveform with noise (c) The waveform with shifted pitch

Fig. 2. An illustration of the data augmentation techniques used.

Data Augmentation - Noise

添加噪音

- 加性高斯白噪聲 (Additive White Gaussian Noise, AWGN)
- 現實噪音由多個源頭的噪音組成
- 中央極限定理說這像是高斯分佈

```
def noise(data):
    noise_amp = 0.035 * np.random.uniform() * np.amax(data)
    data = data + noise_amp * np.random.normal(size=data.shape[0])
    return data
```

Data Augmentation - Pitch

改變音高

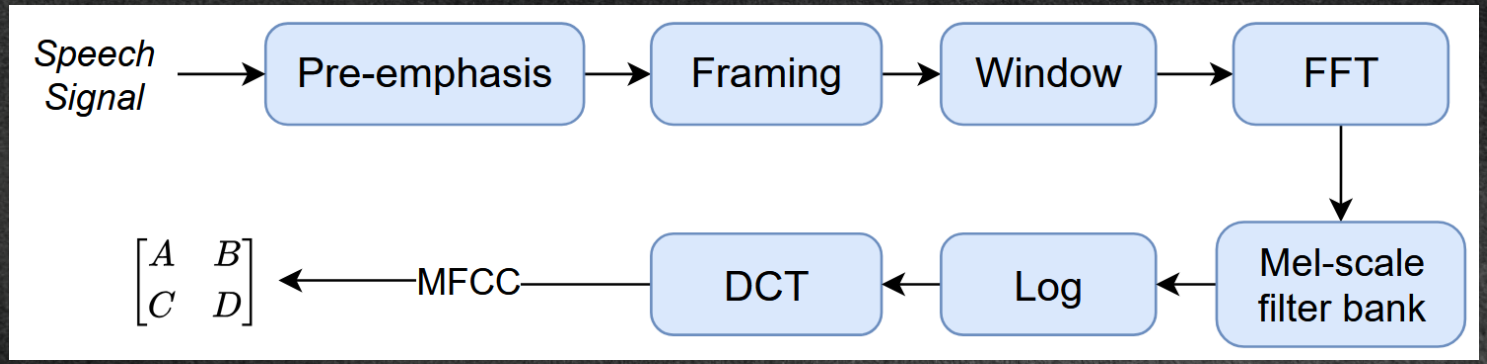
- 基頻擾動 (Fundamental Frequency Perturbation)
- 改變音高來模擬不同人的說話方式

```
def pitch(data, sampling_rate, pitch_factor=0.7):
    return librosa.effects.pitch_shift(data, sr=sampling_rate, n_steps=pitch_factor)
```

built-in

Feature Extraction

- MFCC，對低頻的聲音變化比較敏感，對高頻比較不敏感
- 模擬人類的聽覺特性



Classifier Model

neutral

calm

happy

sad

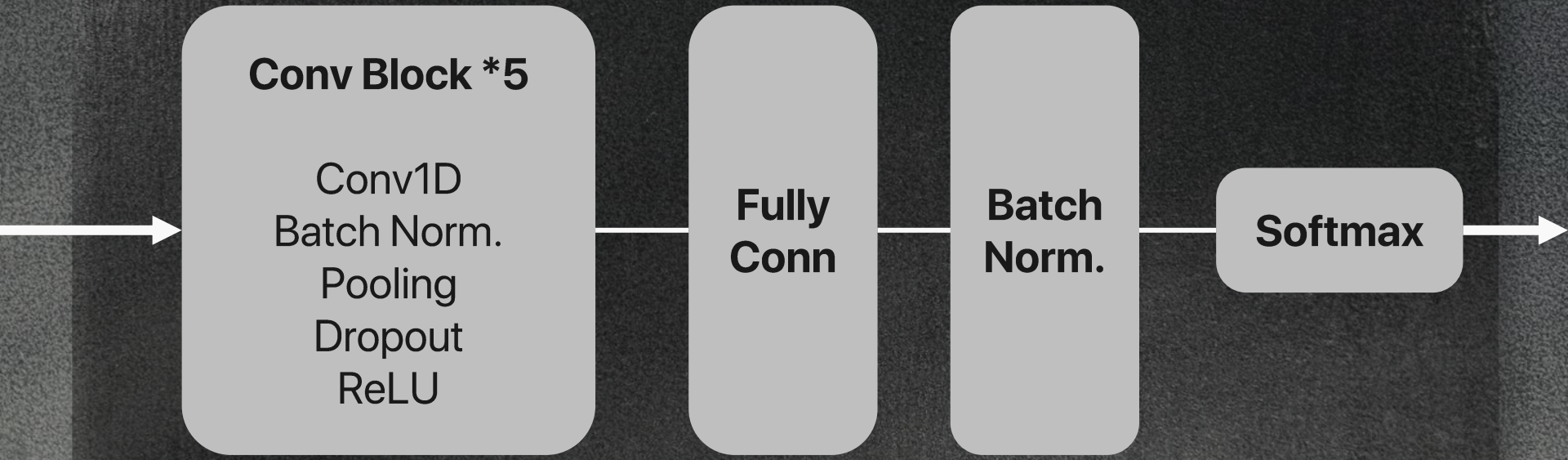
angry

fearful

disgust

surprised

Classifier Model



Classifier Model - 1D CNN

Layer	Number of Filters	Kernel size	Strides	Padding	Activation Function
1	512	5	1	same	ReLU
2	512	5	1	same	ReLU
3	256	5	1	same	ReLU
4	256	3	1	same	ReLU
5	128	3	1	same	ReLU

Training

- 80% 訓練、20% 測試
- 隨機切分

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

PCG64

- 分層抽樣

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

Fisher-Yates Shuffle

Training

- 分層抽樣：80% 訓練、20% 測試
- Optimizer：Adam
  - 學習率 0.001、batch size = 64、Weight Decay = 5 * 10^-4
- Loss Function：cross-entropy (50 iterations)
- 透過 Adam 不斷調整 CNN 裡的參數，讓差距越小越好

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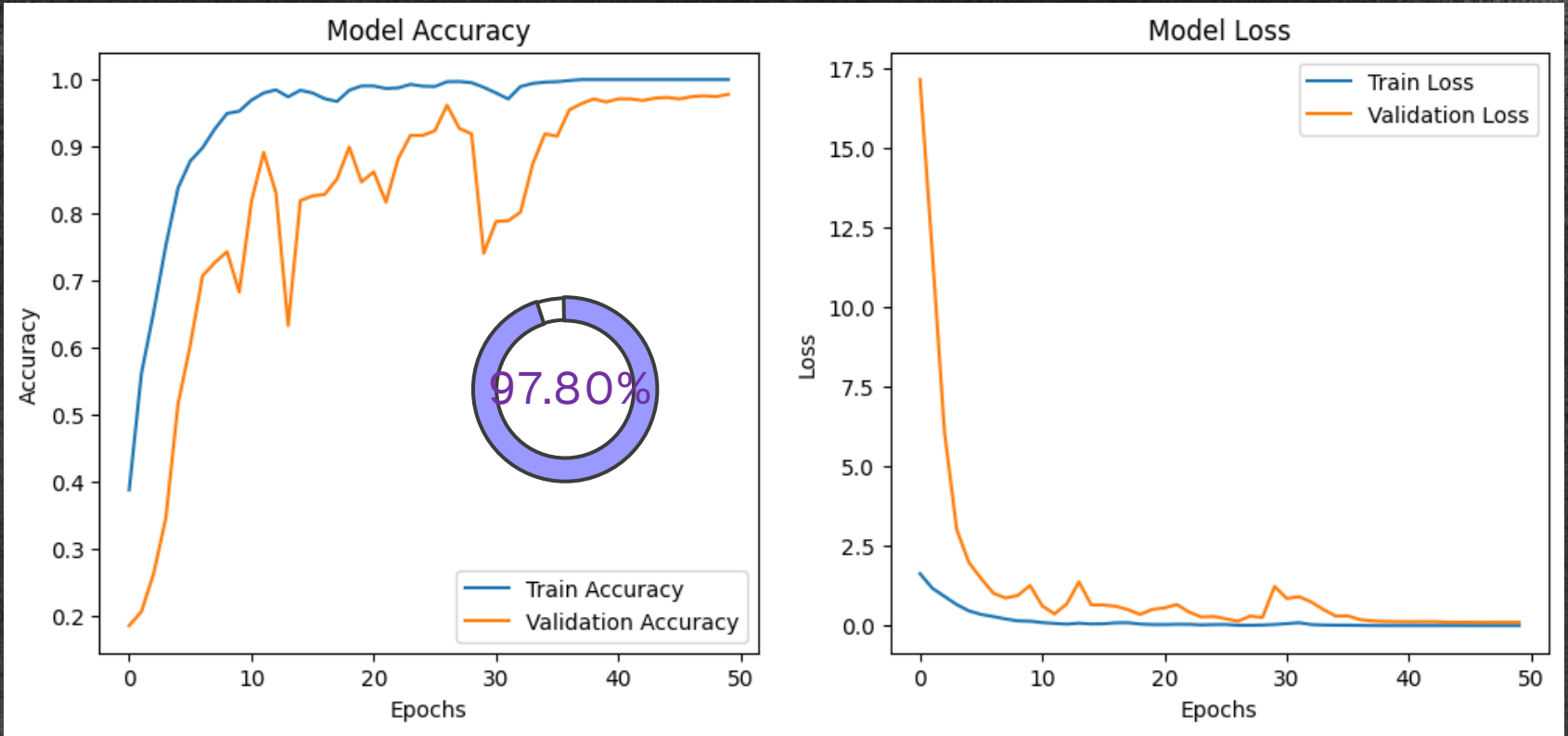
Evaluation

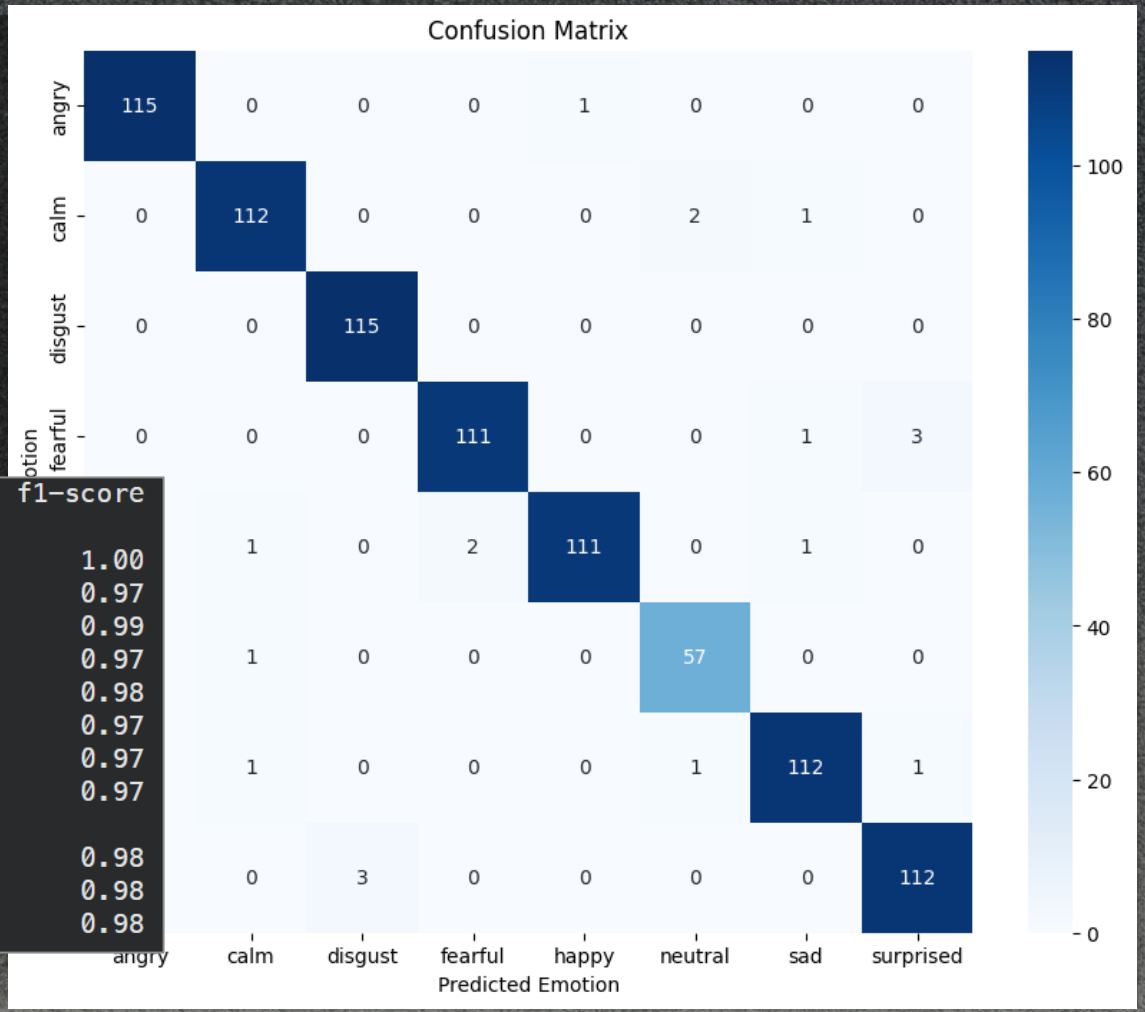
訓練結果評估

Accuracy / Loss (RAVDESS)

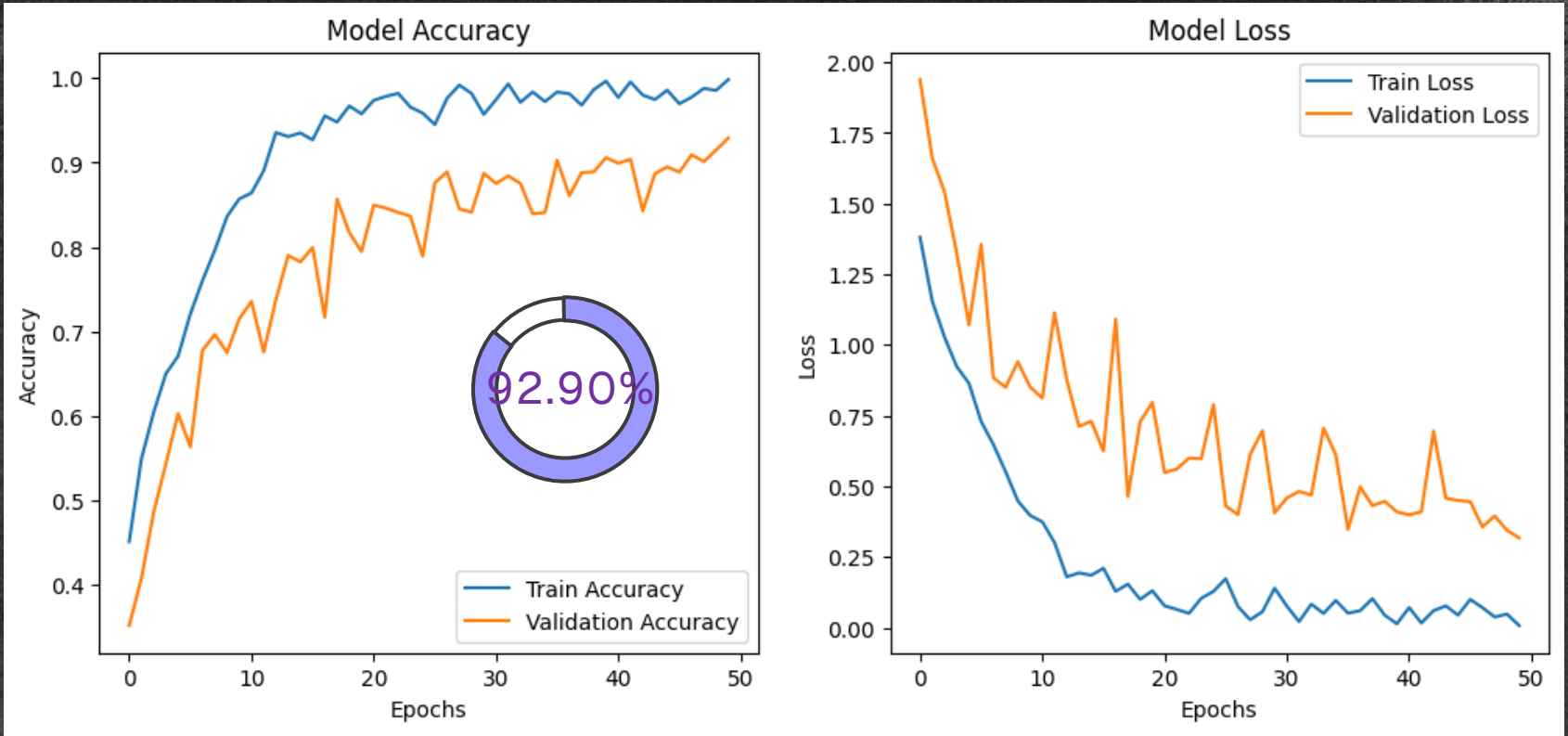


Confusion Matrix (RAVDESS)

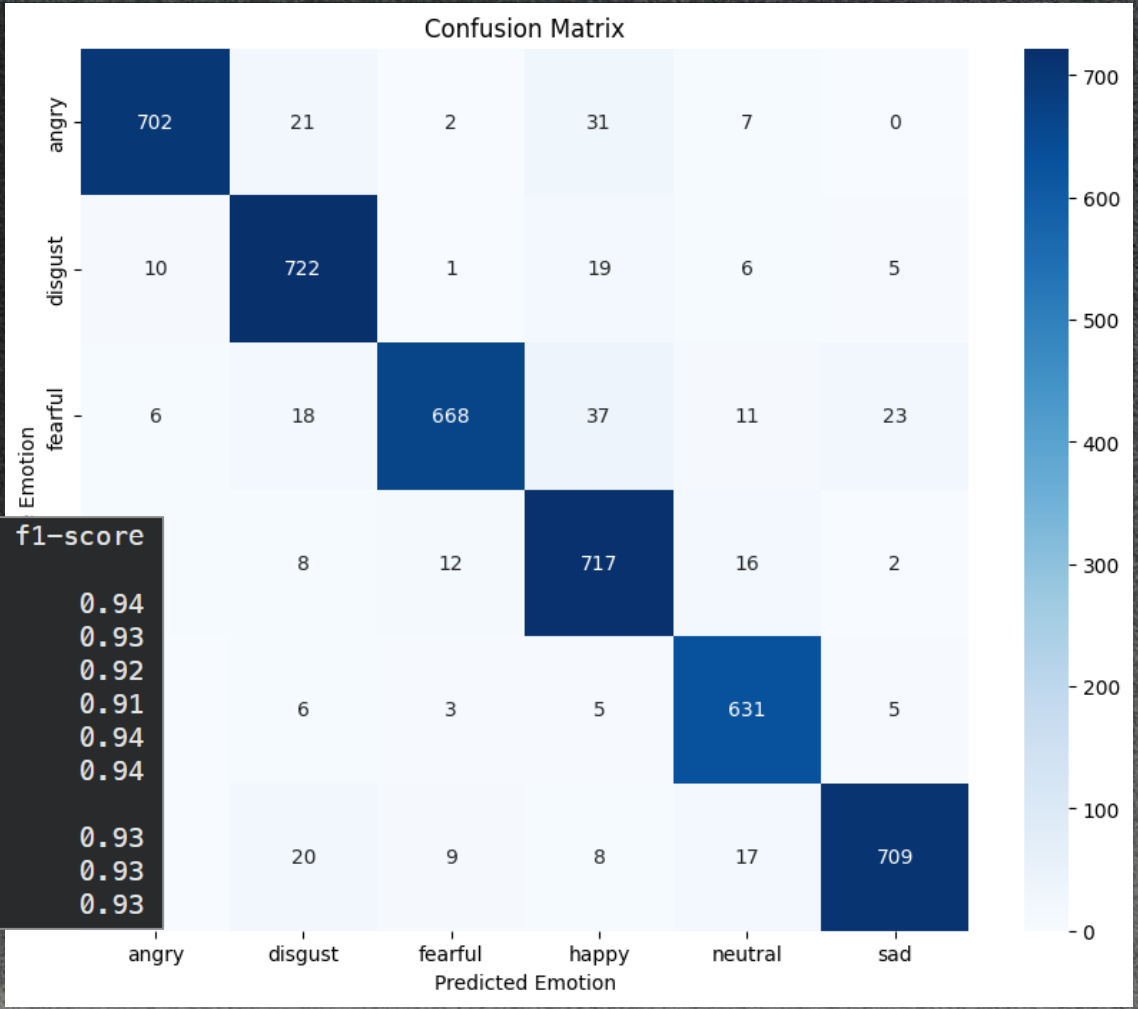
	precision	recall	f1-score
angry	1.00	0.99	1.00
calm	0.97	0.97	0.97
disgust	0.97	1.00	0.99
fearful	0.98	0.97	0.97
happy	0.99	0.97	0.98
neutral	0.95	0.98	0.97
sad	0.97	0.97	0.97
surprised	0.97	0.97	0.97
accuracy			0.98
macro avg	0.98	0.98	0.98
weighted avg	0.98	0.98	0.98



Accuracy / Loss (CREMA-D)



Confusion Matrix (CREMA-D)



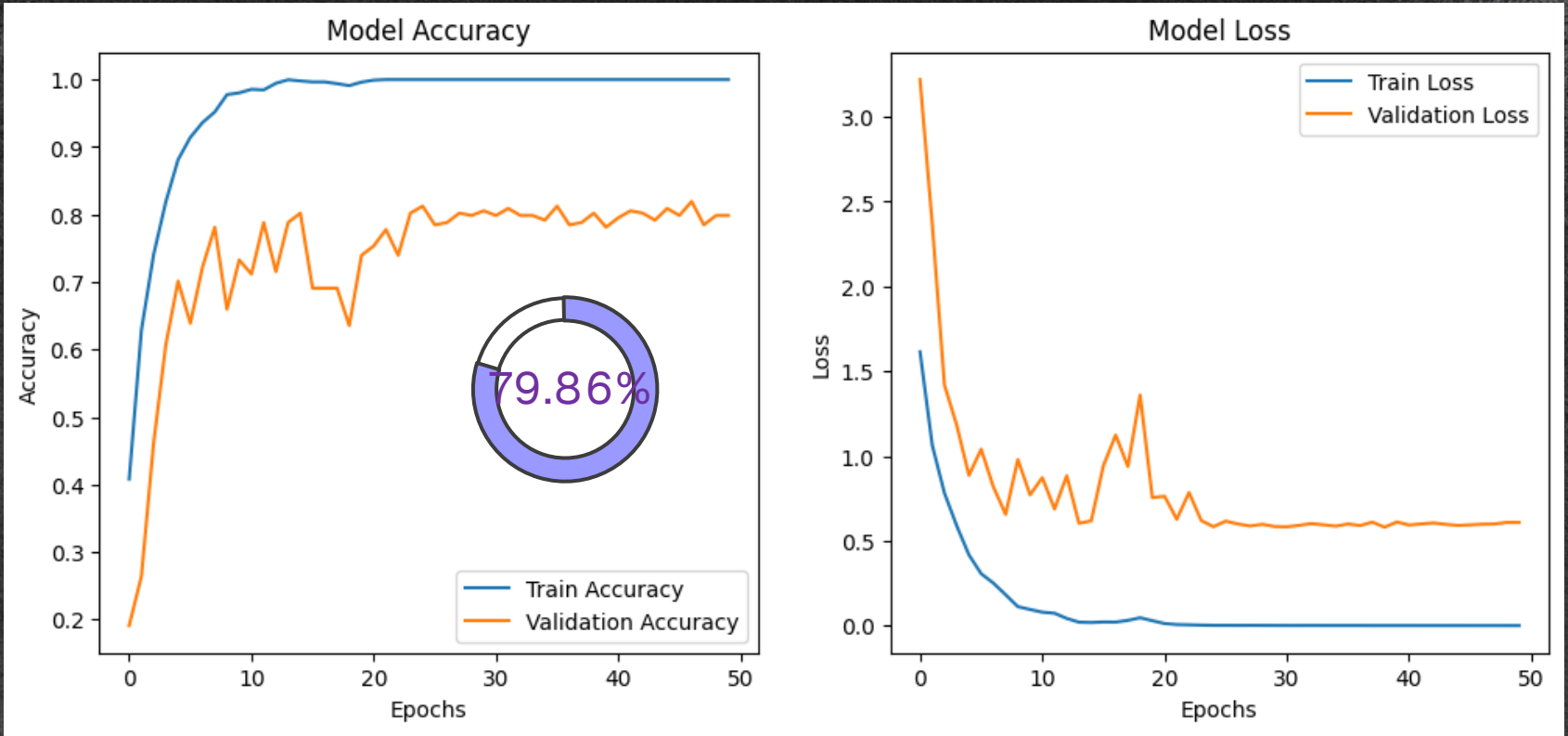
	precision	recall	f1-score
angry	0.97	0.92	0.94
disgust	0.91	0.95	0.93
fearful	0.96	0.88	0.92
happy	0.88	0.94	0.91
neutral	0.92	0.97	0.94
sad	0.95	0.93	0.94
accuracy			0.93
macro avg	0.93	0.93	0.93
weighted avg	0.93	0.93	0.93

Data Leakage

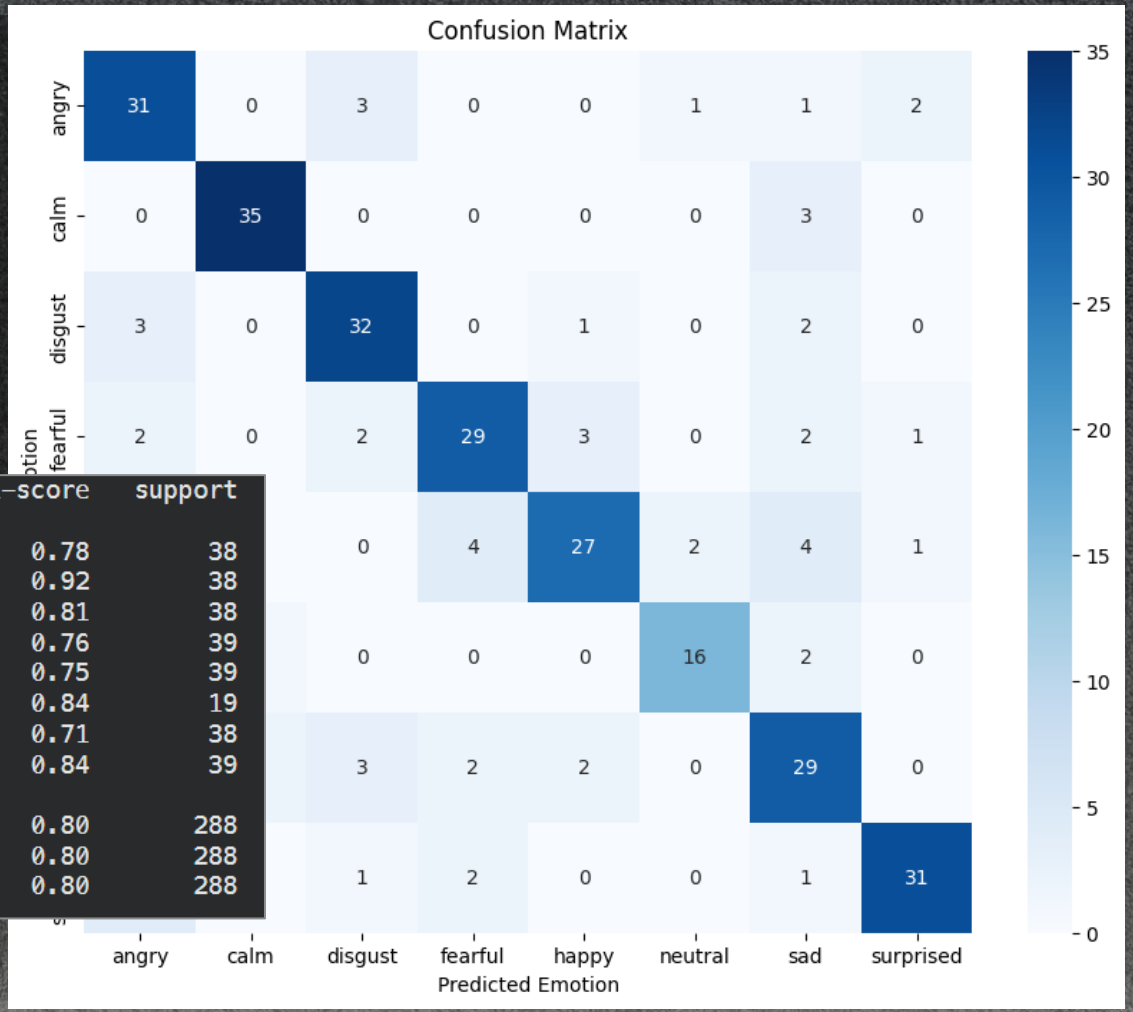
Data Leakage

- 原始訊號與增強訊號間具有極高的 cross-correlation
- 理想情況：驗證集應該與訓練集是統計獨立的
- 驗證集必須是模型從未聽過的聲音
- 只增強訓練集

Accuracy / Loss (RAVDESS)

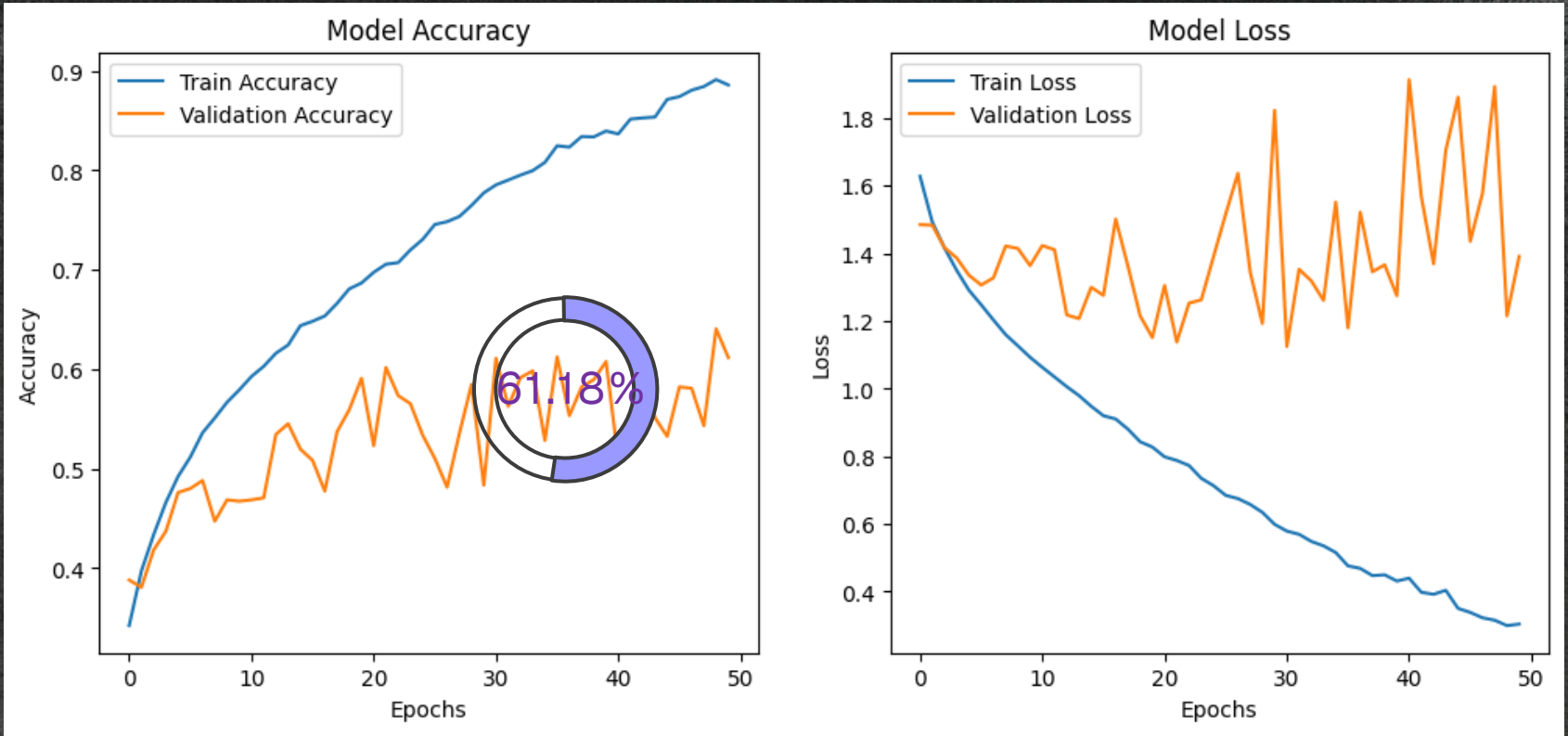


Confusion Matrix (RAVDESS)

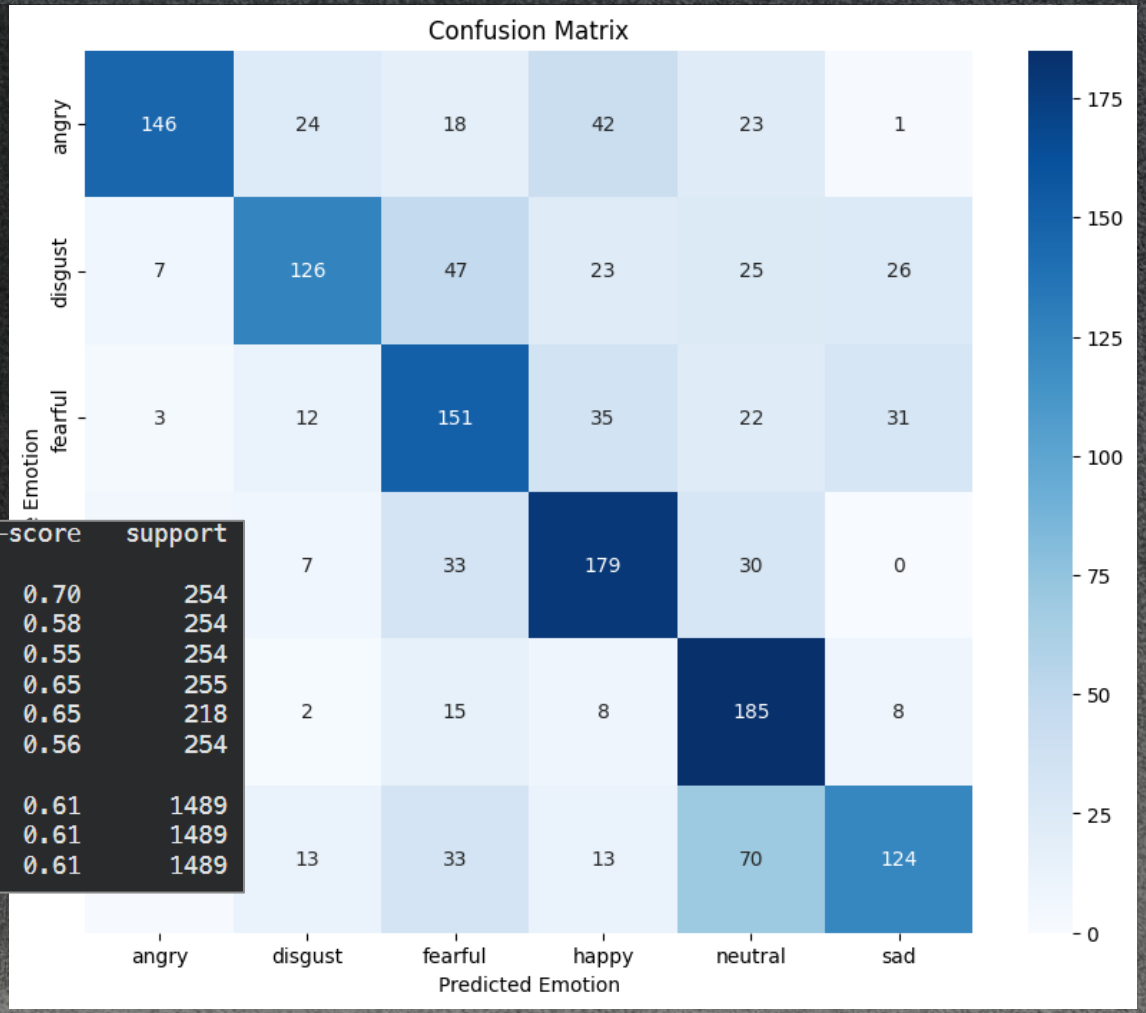


	precision	recall	f1-score	support
angry	0.76	0.82	0.78	38
calm	0.92	0.92	0.92	38
disgust	0.78	0.84	0.81	38
fearful	0.78	0.74	0.76	39
happy	0.82	0.69	0.75	39
neutral	0.84	0.84	0.84	19
sad	0.66	0.76	0.71	38
surprised	0.89	0.79	0.84	39
accuracy			0.80	288
macro avg	0.81	0.80	0.80	288
weighted avg	0.80	0.80	0.80	288

Accuracy / Loss (CREMA-D)



Confusion Matrix (CREMA-D)



	precision	recall	f1-score	support
angry	0.90	0.57	0.70	254
disgust	0.68	0.50	0.58	254
fearful	0.51	0.59	0.55	254
happy	0.60	0.70	0.65	255
neutral	0.52	0.85	0.65	218
sad	0.65	0.49	0.56	254
accuracy			0.61	1489
macro avg	0.64	0.62	0.61	1489
weighted avg	0.65	0.61	0.61	1489

04_

Conclusion

結論

Conclusion

- Python 太開心了，幾乎不用管型別，有很多好用函式
- 了解怎麼對音訊做資料增強
- 資料增強會造成 data leakage
- 認識 1D CNN 的應用

References

- RAVDESS Dataset
- CREMA-D Dataset
- TensorFlow Keras、Scikit-learn
- NumPy、Pandas、librosa、Matplotlib、Python
- Gemini、Codex、Nyquist–Shannon sampling theorem